Chapter 6 HL : Partial fractions and other functions

Mme Doux

I - Partial fractions

Investigation

Show that $\frac{7x+1}{x^2+x-6} \equiv \frac{3}{x-2} + \frac{4}{x+3}$.

a - Methods

Property

Step 1: Factorise the denominator.

Step 2: Write a sum of partial fractions with the factors as denominators and unknown numerators.

Step 3: Multiply both sides by the denominator.

Step 4: Find the unknown numerators by substituting the zeros of the denominators.

The degree of the numerator must be less than the degree of the denominator.

Example

Write as a sum of partial fractions:

$$\frac{7x+1}{x^2+x-6}.$$

Example ⋆

Write

$$\frac{33-12x}{(x+1)(x-2)^2} \quad \text{as the sum of partial fractions in the form} \quad \frac{A}{x+1}+\frac{B}{x-2}+\frac{C}{(x-2)^2}.$$

Remark (⋆). For a repeated factor, substituting the zeros only gives A and C. Find B by comparing the coefficients of x^2 (or by substituting another value of x).

b - Exercises Set I

Exercise 1. Show that:

1. $\frac{3}{x^2+5x+4}$ can be written as the sum of partial fractions $\frac{-1}{x+4}+\frac{1}{x+1}$
2. $\frac{9-x}{x^2+3x-10}$ can be written as the sum of partial fractions $\frac{-2}{x+5}+\frac{1}{x-2}$
3. $\frac{3x-23}{2x^2-5x-12}$ can be written as the sum of partial fractions $\frac{5}{2x+3}-\frac{1}{x-4}$

Exercise 2. Write as the sum of partial fractions:

1. $\frac{1}{x^2+3x+2}$
2. $\frac{x}{x^2-5x+6}$
3. $\frac{3x}{x^2-2x-8}$
4. $\frac{3x+2}{x^2-4}$
5. $\frac{2x-1}{x^2-2x-3}$
6. $\frac{2-5x}{x^2+4x-12}$
7. $\frac{7x+10}{2x^2+5x+3}$
8. $\frac{x+9}{3x^2-5x-2}$
9. $\frac{2x+27}{8x^2+18x-5}$

Exercise 3 (⋆). Show that

$$\frac{5x^2+11x+16}{(x-5)(x+2)^2} \quad \text{can be written as the sum of partial fractions} \quad \frac{4}{x-5}+\frac{1}{x+2}-\frac{2}{(x+2)^2}.$$

Exercise 4 (⋆). Write

$$\frac{-3x-4}{(x+2)(x+3)^2} \quad \text{as the sum of partial fractions in the form} \quad \frac{A}{x+2}+\frac{B}{x+3}+\frac{C}{(x+3)^2}.$$

Exercise 5 ($\star$).
1. Show that $(x+3)(x-1)^2 = x^3 + x^2 - 5x + 3$.
2. Write $\dfrac{19x+9}{x^3+x^2-5x+3}$ as the sum of partial fractions.

Exercise 6 ($\star$).
1. Show that $(2x-1)(x^2+x+4) = 2x^3 + x^2 + 7x - 4$.
2. Write
$$\frac{10x^2-3x+18}{2x^3+x^2+7x-4} \quad \text{in the form} \quad \frac{A}{2x-1} + \frac{Bx+C}{x^2+x+4}.$$

II - Rational functions

a - Asymptotes

Investigation

For a function of the form
$$y = \frac{ax+b}{cx^2+dx+e}, \quad c \neq 0 :$$
1. What is the horizontal asymptote?
2. How can we determine the vertical asymptotes?

Property

For a function of the form
$$y = \frac{ax+b}{cx^2+dx+e}, \quad c \neq 0 :$$
- The horizontal asymptote is $y = 0$.
- The vertical asymptotes are the zeros of the denominator cx^2+dx+e (a zero of the denominator which is also a zero of the numerator gives a hole, not an asymptote).

Example

Consider the function

$$y = \frac{2x - 6}{x^2 - 3x - 4}.$$

- Find the asymptotes of the function.
- Find the axes intercepts.
- Draw a sign diagram of the function.
- Hence discuss the behaviour of the function near the asymptotes.
- Sketch the function, showing the features you have found.

Investigation

Consider the function

$$y = \frac{6x^2 + x - 3}{2x + 1}.$$

1. State the equation of the vertical asymptote.

2. Show that

$$\frac{6x^2 + x - 3}{2x + 1} = 3x - 1 - \frac{2}{2x + 1}.$$

3. Use technology to sketch

$$y = \frac{6x^2 + x - 3}{2x + 1} \quad \text{and} \quad y = 3x - 1$$

on the same set of axes.

4. Describe the behaviour of

$$y = \frac{6x^2 + x - 3}{2x + 1}$$

as $|x| \to \infty$.

Property

To help us describe graphs of the form

$$y = \frac{ax^2 + bx + c}{dx + e}, \quad a \neq 0,\ d \neq 0,$$

we use polynomial division to write them in the form

$$y = px + q + \frac{r}{dx + e}.$$

- The function is undefined when $dx + e = 0$, so (if $r \neq 0$) the graph has the vertical asymptote

$$x = -\frac{e}{d}.$$

- As $|x| \to \infty$, the term

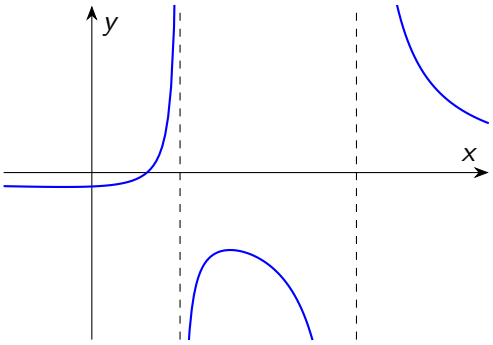
$$\frac{r}{dx + e} \to 0,$$

and so the graph gets closer to the line $y = px + q$. The graph has the **oblique asymptote** $y = px + q$.

Remark. An oblique asymptote is an asymptote which is neither vertical nor horizontal.

b - Exercises Set II

Exercise 7. The graph of $y = \dfrac{4x-5}{x^2-8x+12}$ is shown below.



a) Find the asymptotes of the function.

b) Find the axes intercepts.

Exercise 8. For each of the following functions:

i. Find the equations of the asymptotes.

ii. Find the axes intercepts.

iii. Draw a sign diagram of the function.

iv. Hence discuss the behaviour of the function near its asymptotes.

v. Sketch the graph of the function.

a) $y = \dfrac{x-1}{(x+3)(x-2)}$

b) $y = \dfrac{2x-8}{(x-1)(x-3)}$

c) $f(x) = \dfrac{5-3x}{(x+2)(x-4)}$

d) $f(x) = \dfrac{-x-4}{x^2+3x}$

e) $y = \dfrac{2x+5}{x^2-4x-5}$

f) $y = \dfrac{12-4x}{x^2-3x-18}$

g) $f(x) = \dfrac{5x-9}{2x^2+10x-12}$

h) $y = \dfrac{3x+6}{x^2+2x+1}$

i) $f(x) = \dfrac{4-x}{2x^2-9x-5}$

j) $y = \dfrac{6x+15}{3x^2+2x-8}$

k) $f(x) = \dfrac{7-4x}{21+8x-4x^2}$

l) $y = \dfrac{1-3x}{9x^2-12x+4}$

Exercise 9. Consider the function $y = \dfrac{4x-6}{x^2-5x+7}$.

1. State the equation of the horizontal asymptote.

2. Explain why the function has no vertical asymptotes.

3. Find the axes intercepts.

4. Use technology to help sketch the function.

Exercise 10. The function $f(x) = \dfrac{8x-3}{cx^2+dx+9}$ has one vertical asymptote, and passes through the point $(1, 5)$.

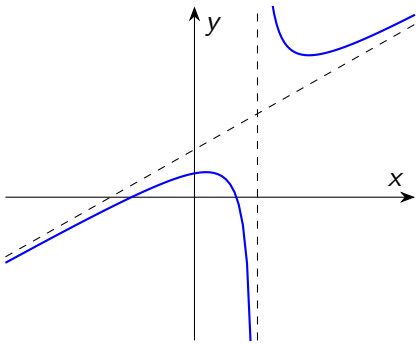
1. Show that $f(x)$ can take two possible forms. Find the vertical asymptote of $f(x)$ in each case.
2. Sketch the possible graphs of $y = f(x)$ on the same set of axes.

Exercise 11. The graph of $y = \dfrac{3x^2+x-2}{x-1}$ is shown below.

a) Find the equation of the vertical asymptote.

b) Find the axes intercepts.

c) Write the function in the form $y = px+q+\dfrac{r}{x-1}$, and hence find the equation of the oblique asymptote.



Exercise 12. For each of the following functions:

i. Find the equation of the vertical asymptote.

ii. Find the axes intercepts.

iii. Find the oblique asymptote.

iv. Draw a sign diagram of the function.

v. Hence discuss the behaviour of the function near its asymptotes.

vi. Sketch the graph of the function.

a) $y = \dfrac{x^2+x-6}{x-1}$

b) $f(x) = \dfrac{-x^2+3x-2}{x}$

c) $y = \dfrac{2x^2-8x+8}{x-3}$

d) $f(x) = \dfrac{-6x^2+5x}{2x+1}$

e) $y = \dfrac{-6x^2-4x-1}{3x+2}$

f) $f(x) = \dfrac{8x^2-19x-15}{1-2x}$

III - Even and odd functions

a - Definitions and examples

Definition

A function $f(x)$ is **even** if $f(-x) = f(x)$ for all x in the domain of f.

The graph of an even function is symmetrical about the y-axis.

Definition

A function $f(x)$ is **odd** if $f(-x) = -f(x)$ for all x in the domain of f.

The graph of an odd function has rotational symmetry of order 2 about the origin.

Remark. For f to be even or odd, its domain must be symmetric about 0 (if x is in the domain, so is $-x$).

Example

Determine whether each function is even, odd, or neither:

1. $f(x) = \dfrac{1}{x}$
2. $g(x) = x^4 + x^2$
3. $h(x) = x^2 - 2x + 3$

Example

Prove that the sum of two odd functions is also an odd function.

b - Exercises Set III

Exercise 13. Determine whether each function is even, odd, or neither:

a) $f(x) = 5x$

b) $f(x) = -4x + 3$

c) $f(x) = \dfrac{3}{x^2 - 4}$

d) $f(x) = 2x^3 - \dfrac{5}{x}$

e) $f(x) = x^2 + \dfrac{7}{x^2} - 3$

f) $f(x) = \sqrt{x}$

Exercise 14. Suppose $g(x)$ is an odd function, and $(4, 6)$ and $(-1, 2)$ lie on the graph of $y = g(x)$. Write down two other points which lie on the graph of $y = g(x)$.

Exercise 15. Suppose $f(x) = (2x + 3)(x + a)$, $a \in \mathbb{R}$, is an even function. Find the value of a.

Exercise 16. a) If the quadratic function $f(x) = ax^2 + bx + c$, $a \neq 0$, is an even function, prove that $b = 0$.

b) Under what conditions is the cubic function $g(x) = ax^3 + bx^2 + cx + d$, $a \neq 0$, an odd function?

c) Under what conditions is the quartic function $h(x) = ax^4 + bx^3 + cx^2 + dx + e$, $a \neq 0$, an even function?

Exercise 17. Decide whether each of the following is even, odd, or neither. Explain how this is observed in the graph of the function.

a) $f(x) = \cos x$

b) $f(x) = \sin x$

c) $f(x) = \tan x$

Exercise 18. For what values of k is $f(x) = \cos(x - k)$:

a) even

b) odd

c) neither?

Exercise 19. Prove that:

a) the sum of two even functions is an even function

b) the difference between two odd functions is an odd function

c) the product of two odd functions is an even function.

Exercise 20. Suppose $f(x)$ is an even function and $g(x)$ is an odd function. Determine whether the function $(f \circ g)(x)$ is even, odd, or neither.

IV - Squared functions

a - Intercepts, asymptotes and intersections

Investigation

1. Sketch each pair of functions on the same set of axes. Include all axes intercepts and asymptotes.

   (a) $y = x + 1$ and $y = (x + 1)^2$

   (b) $y = x^2 - 4$ and $y = (x^2 - 4)^2$

   (c) $y = \dfrac{2}{x}$ and $y = \left(\dfrac{2}{x}\right)^2$

   (d) $y = \dfrac{1}{x - 3}$ and $y = \dfrac{1}{(x - 3)^2}$

   (e) $y = 2^x - 1$ and $y = (2^x - 1)^2$

   (f) $y = \log x$ and $y = (\log x)^2$

2. Determine the points where each pair of functions intersect. What do you notice?

3. What other observations can you make about the graph of $y = [f(x)]^2$?

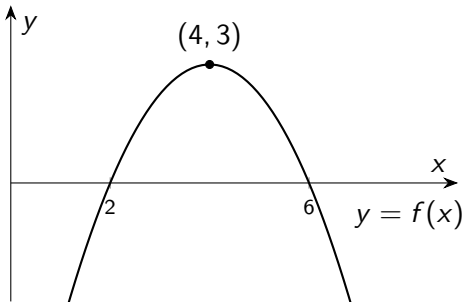
Property

When $y = [f(x)]^2$ is graphed from $y = f(x)$:

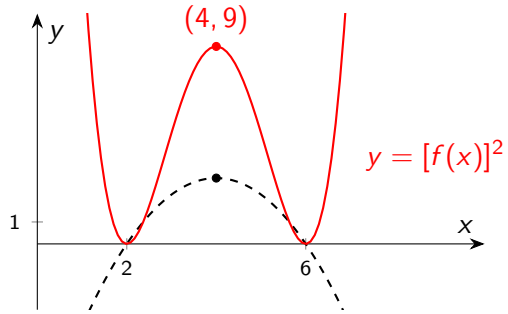
- The points with y-coordinate 0 or 1 are invariant.
- The graph of $y = [f(x)]^2$ touches the x-axis at its x-intercepts.
- The graph of $y = [f(x)]^2$ lies above or on the x-axis for all x.
- The vertical asymptotes of $y = f(x)$ are also vertical asymptotes of $y = [f(x)]^2$.
- A turning point (a, b) of $y = f(x)$ gives a turning point (a, b^2) of $y = [f(x)]^2$.
- When $-1 < f(x) < 1$, $y = [f(x)]^2$ is closer to the x-axis than $y = f(x)$.
- When $|f(x)| > 1$, $y = [f(x)]^2$ is further from the x-axis than $y = f(x)$.

Example

For the graph of $y = f(x)$ shown, draw the graph of $y = [f(x)]^2$.

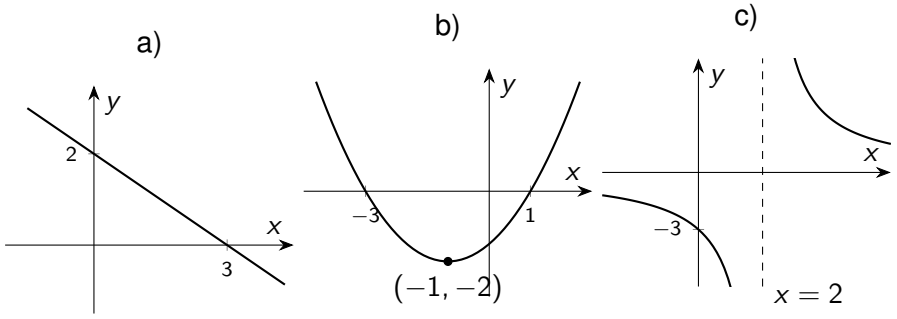


Solution. $y = f(x)$ cuts the x-axis at 2 and 6, so $y = [f(x)]^2$ touches the x-axis at 2 and 6. The graphs also intersect when $y = 1$. $y = f(x)$ has a turning point at $(4, 3)$, so $y = [f(x)]^2$ has a turning point at $(4, 9)$.



b - Exercises Set IV

Exercise 21. For each graph of $y = f(x)$, sketch the graph of $y = [f(x)]^2$.



Exercise 22. a) Draw the graph of $y = f(x) = 3^x$.

b) On the same set of axes, sketch the graph of $y = [f(x)]^2$.

c) Describe a stretch which transforms $y = f(x)$ to $y = [f(x)]^2$.

Exercise 23. a) Draw the graph of $y = f(x) = \log_3 x$.

b) On the same set of axes, sketch the graph of $y = [f(x)]^2$.

c) Find the invariant points when $y = f(x)$ is transformed to $y = [f(x)]^2$.

Exercise 24. Show that if $f(x)$ is odd, then $[f(x)]^2$ is even.

V - Absolute value functions

a - Graph

Definition

The **absolute value** or **modulus** of a real number x is its distance from 0 on the number line. We write the absolute value of x as $|x|$.

Remark. Because the absolute value is a distance, it cannot be negative.

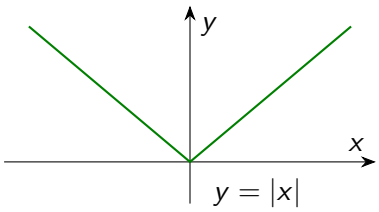
Property

The absolute value of x is

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0, \end{cases}$$

or alternatively

$$|x| = \sqrt{x^2}.$$



Property

To obtain the graph of $y = |f(x)|$ from the graph of $y = f(x)$:

- keep the graph for $f(x) \geq 0$
- reflect the graph in the x-axis for $f(x) < 0$, discarding what was there
- points on the x-axis are invariant.

Remark. **Invariant** points do not move under a transformation.

Property

To obtain the graph of $y = f(|x|)$ from the graph of $y = f(x)$:

- discard the graph for $x < 0$
- reflect the graph for $x \geq 0$ in the y-axis, keeping what was there
- points on the y-axis are invariant.

Example

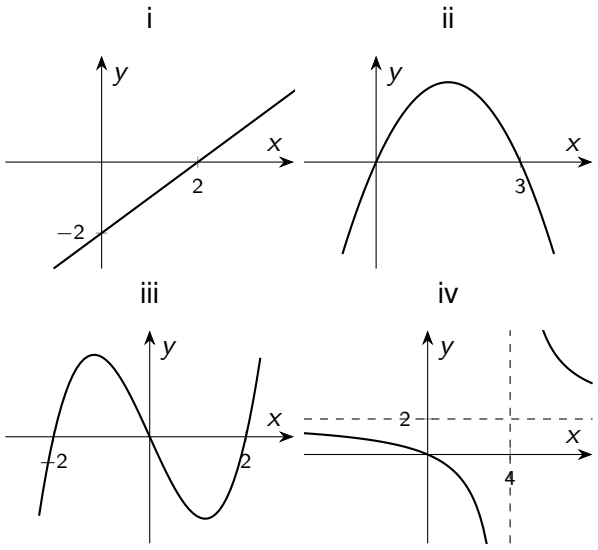
Draw the graph of $f(x) = 3x(x-2)$, and on the same set of axes draw the graph of:

(a) $y = |f(x)|$

(b) $y = f(|x|)$

Exercise 25. a) Copy the following graphs of $y = f(x)$ and, on the same set of axes, draw the graph of $y = |f(x)|$.

b) Repeat a), but this time draw the graphs of $y = f(x)$ and $y = f(|x|)$ on the same set of axes.



Exercise 26. Let $f(x) = \sqrt{4-x}$.

a) Draw the graph of $y = f(x)$ and state its domain and range.

b) Draw the graph of $y = f(|x|)$ and state its domain and range.

Exercise 27. Use the definition $|x| = \sqrt{x^2}$ to prove that:

a) $|-x| = |x|$ for all x

b) $|x|^2 = x^2$ for all x

c) $|xy| = |x||y|$ for all x, y

d) $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$ for all x and y, $y \neq 0$

e) $|x-y| = |y-x|$ for all x, y

Example

Use the definition

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0, \end{cases}$$

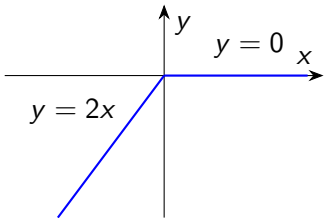
to write the following functions without the modulus sign. Hence graph each function:

(a) $f(x) = x - |x|$

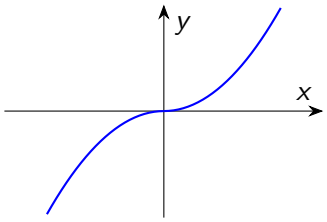
(b) $f(x) = x|x|$

SOLUTION

(a) If $x < 0$, $f(x) = x - (-x) = 2x$.
If $x \geq 0$, $f(x) = x - x = 0$.



(b) If $x \geq 0$, $f(x) = x \cdot x = x^2$.
If $x < 0$, $f(x) = x(-x) = -x^2$.



Exercise 28. Use $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ to write the following functions without modulus signs. Hence graph each function.

a) $y = |x-2|$

b) $y = |x+1|$

c) $y = -|x|$

d) $y = |x| + x$

e) $y = \dfrac{|x|}{x}$

f) $y = x - 2|x|$

g) $y = |x| + |x-2|$

h) $y = |x| - |x-1|$

b - Modulus equations and inequalities

Property

- $|x| \geq 0$ for all x.
- $|x|^2 = x^2$ for all x.
- $\left|\dfrac{x}{y}\right| = \dfrac{|x|}{|y|}$ for all x and y, $y \neq 0$.
- $|-x| = |x|$ for all x.
- $|xy| = |x||y|$ for all x and y.
- $|x-y| = |y-x|$ for all x and y.

Property

If $|x| = a$ where $a > 0$, then $x = \pm a$.

If $a < 0$, $|x| = a$ has no solution.

Proof

If $|x| = a$, $a > 0$, then

$$|x|^2 = a^2$$
$$\therefore\ x^2 = a^2 \qquad \{|x|^2 = x^2\}$$
$$\therefore\ x = \pm\sqrt{a^2}$$
$$\therefore\ x = \pm a.$$

Example

Solve for x:

(a) $|2x+3| = 7$

(b) $|3-2x| = -1$

SOLUTION

(a) $|2x+3| = 7$

$\therefore\ 2x+3 = \pm 7$

$\therefore\ 2x+3 = 7 \quad \text{or} \quad 2x+3 = -7$

$\therefore\ 2x = 4 \quad \text{or} \quad 2x = -10$

$\therefore\ x = 2 \quad \text{or} \quad x = -5.$

(b) $|3-2x| = -1$ has no solution, as the left-hand side is never negative.

Example

Solve for x:

(a) $\left|\dfrac{3x+2}{1-x}\right| = 4$

(b) $|x+1| = |2x-3|$

SOLUTION

(a) If $\left|\dfrac{3x+2}{1-x}\right| = 4$, then $\dfrac{3x+2}{1-x} = \pm 4$ and $x \neq 1$.

$3x+2 = 4(1-x) \quad \text{or} \quad 3x+2 = -4(1-x)$

$3x+2 = 4-4x \quad \text{or} \quad 3x+2 = -4+4x$

$7x = 2 \quad \text{or} \quad 6 = x$

$x = \dfrac{2}{7} \quad \text{or} \quad x = 6.$

(b) If $|x+1| = |2x-3|$, then $x+1 = \pm(2x-3)$.

$x+1 = 2x-3 \quad \text{or} \quad x+1 = -(2x-3)$

$4 = x \quad \text{or} \quad x+1 = -2x+3$

$\text{or} \quad 3x = 2$

$x = 4 \quad \text{or} \quad x = \dfrac{2}{3}.$

Exercise 29. Solve for x:

a) $|x| = 3$

b) $|x| = -5$

c) $|x| = 0$

d) $|x-1| = 3$

e) $|3-x| = 4$

f) $|x+5| = -1$

Exercise 30. Solve for x:

a) $\left|\dfrac{x}{x-1}\right| = 3$

b) $\left|\dfrac{2x-1}{x+1}\right| = 5$

c) $\left|\dfrac{x+3}{1-3x}\right| = \dfrac{1}{2}$

Exercise 31. Solve for x:

a) $|3x-1| = |x+2|$

b) $|2x+5| = |1-x|$

c) $|x+1| = |2-x|$

d) $|x| = |5-x|$

e) $|1-4x| = 2|x-1|$

f) $|3x+2| = 2|2-x|$

Exercise 32. Solve graphically (use technology):

a) $|x+2| = 2x+1$

b) $|2x+3| = 3|x|$

c) $|x-2| = \frac{2}{5}x+1$

d) $x^2-1 = |5x-x^2|$

e) $|e^x-6| = 2^x$

f) $x^3-4x = \ln|x|$

Example

Solve for x:

(a) $|2x-1| < 3$

(b) $|4-x| \geq 5$

Example

Solve for x:

(a) $|3x+1| \geq |2x-3|$

(b) $\left|\frac{x}{x+1}\right| \geq \frac{1}{2}$

Exercise 33. Solve for x:

a) $|x+5| < |x-1|$

b) $|2x+1| > |x+2|$

c) $|x-3| \leq |2x-6|$

d) $|3x+4| \geq |x+3|$

e) $2|x-4| < |3x+2|$

f) $\frac{1}{3}|2x+5| \geq |4-x|$

Exercise 34. Solve for x:

a) $\left|\frac{x}{x-2}\right| \geq 3$

b) $\left|\frac{2x+3}{x-1}\right| \geq 2$

c) $\left|\frac{x-4}{1-2x}\right| < \frac{2}{3}$

Exercise 35. Solve graphically (use technology):

a) $|2x-3| < x$

b) $|x|-2 \geq |4-x|$

c) $x^2-4 > |x-1|$

d) $|2x-1|+|x-4| \leq 10-x$